POWER BI WEEK – 9 REPORT

Introduction

Week 9 focuses on Housing Data Cleaning and Transformation using Power BI. The King County House Sales dataset is used to analyze historical house sales data and understand property characteristics and factors associated with house prices.

The activity involves importing and cleaning housing data, handling missing and invalid values, extracting useful information from sale dates, and creating calculated columns such as property age, price per square foot, total area, and renovation status. Properties are also categorized based on size and price ranges.

Overall, this activity aims to prepare a clean and structured housing dataset for further analysis and the development of an interactive Real-Estate Analytics Dashboard in Week 10.

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Figure 1: Importing the King County House Sales dataset into Power BI for data cleaning and transformation.

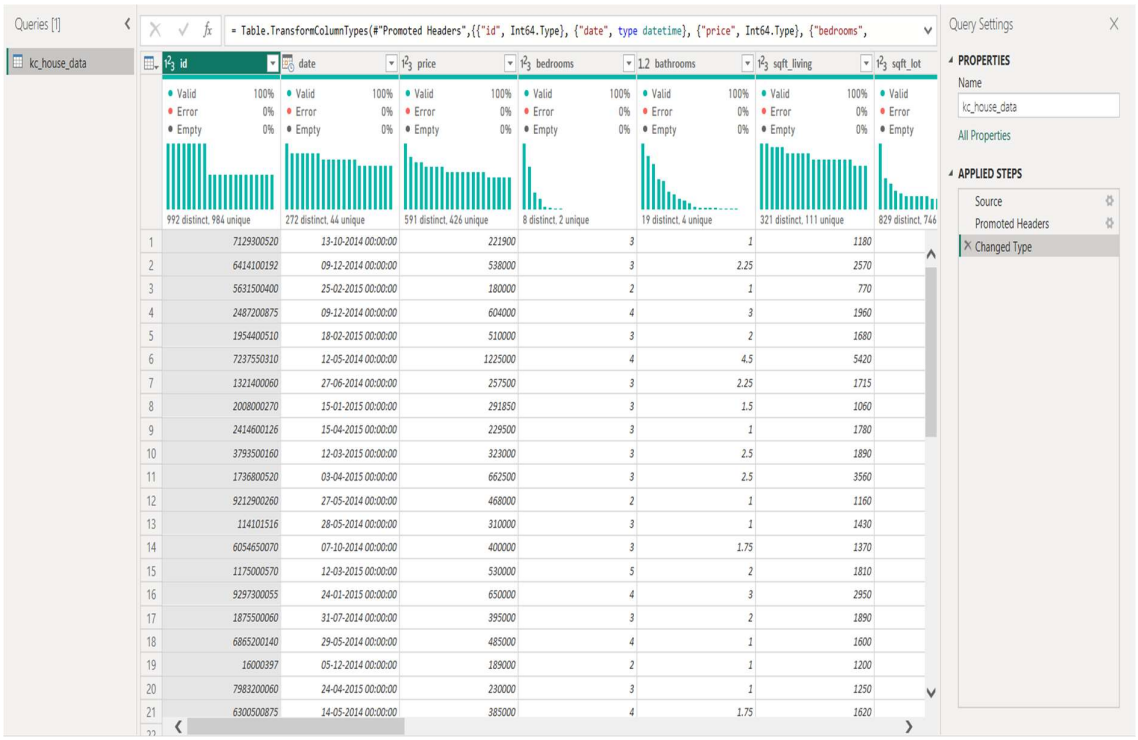
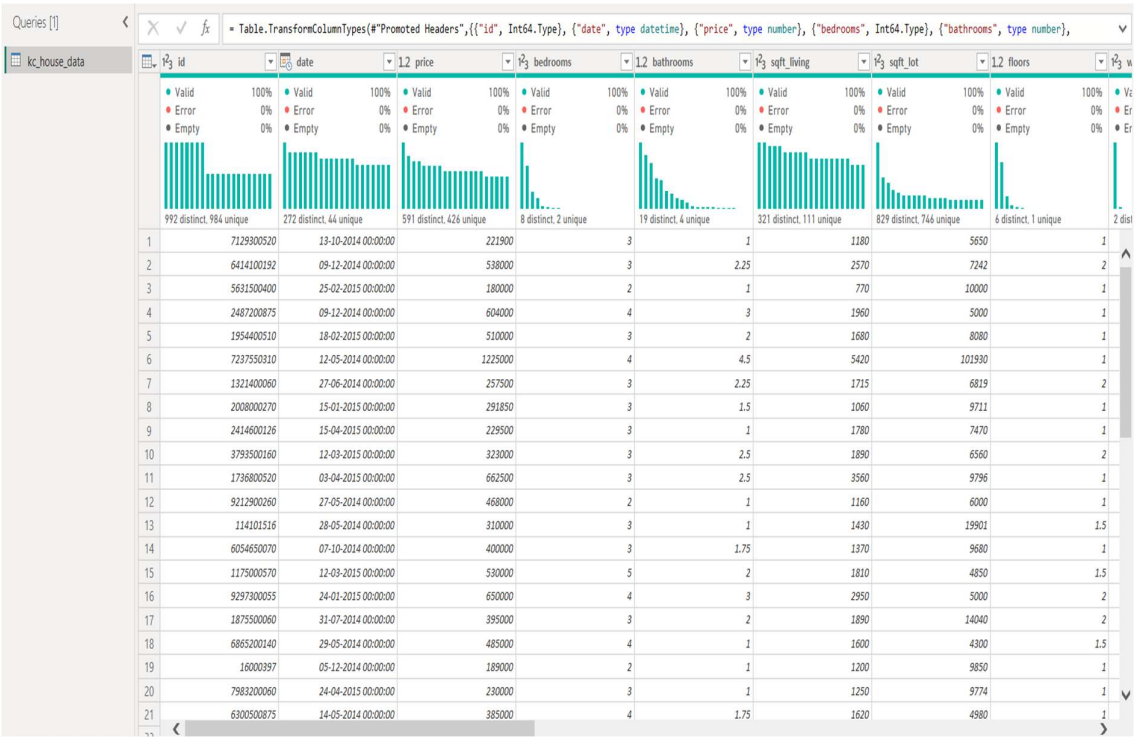


Figure 2: Inspecting all 21 columns and verifying their data types in Power Query.



	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors
1	7129300520	13-10-2014 00:00:00	221900	3	1	1180	5650	1
2	6414100192	09-12-2014 00:00:00	538000	3	2.25	2570	7242	2
3	5631500400	25-02-2015 00:00:00	180000	2	1	770	10000	1
4	2487200875	09-12-2014 00:00:00	604000	4	3	1960	5000	1
5	1954400510	18-02-2015 00:00:00	510000	3	2	1680	8080	1
6	7237550310	12-05-2014 00:00:00	1225000	4	4.5	5420	101930	1
7	1321400060	27-06-2014 00:00:00	257500	3	2.25	1715	6819	2
8	2008000270	15-01-2015 00:00:00	291850	3	1.5	1060	9711	1
9	2414600126	15-04-2015 00:00:00	229500	3	1	1780	7470	1
10	3793500160	12-03-2015 00:00:00	323000	3	2.5	1890	6560	2
11	1736800520	03-04-2015 00:00:00	662500	3	2.5	3560	9796	1
12	9212900260	27-05-2014 00:00:00	468000	2	1	1160	6000	1
13	114101516	28-05-2014 00:00:00	310000	3	1	1430	19901	1.5
14	6054650070	07-10-2014 00:00:00	400000	3	1.75	1370	9680	1
15	1175000570	12-03-2015 00:00:00	530000	5	2	1810	4850	1.5
16	9297300055	24-01-2015 00:00:00	650000	4	3	2950	5000	2
17	1875500060	31-07-2014 00:00:00	395000	3	2	1890	14040	2
18	6865200140	29-05-2014 00:00:00	485000	4	1	1600	4300	1.5
19	16000397	05-12-2014 00:00:00	189000	2	1	1200	9850	1
20	7983200060	24-04-2015 00:00:00	230000	3	1	1250	9774	1
21	6300500875	14-05-2014 00:00:00	385000	4	1.75	1620	4980	1

Figure 3: Checking for duplicate property IDs in Power Query to maintain data quality.

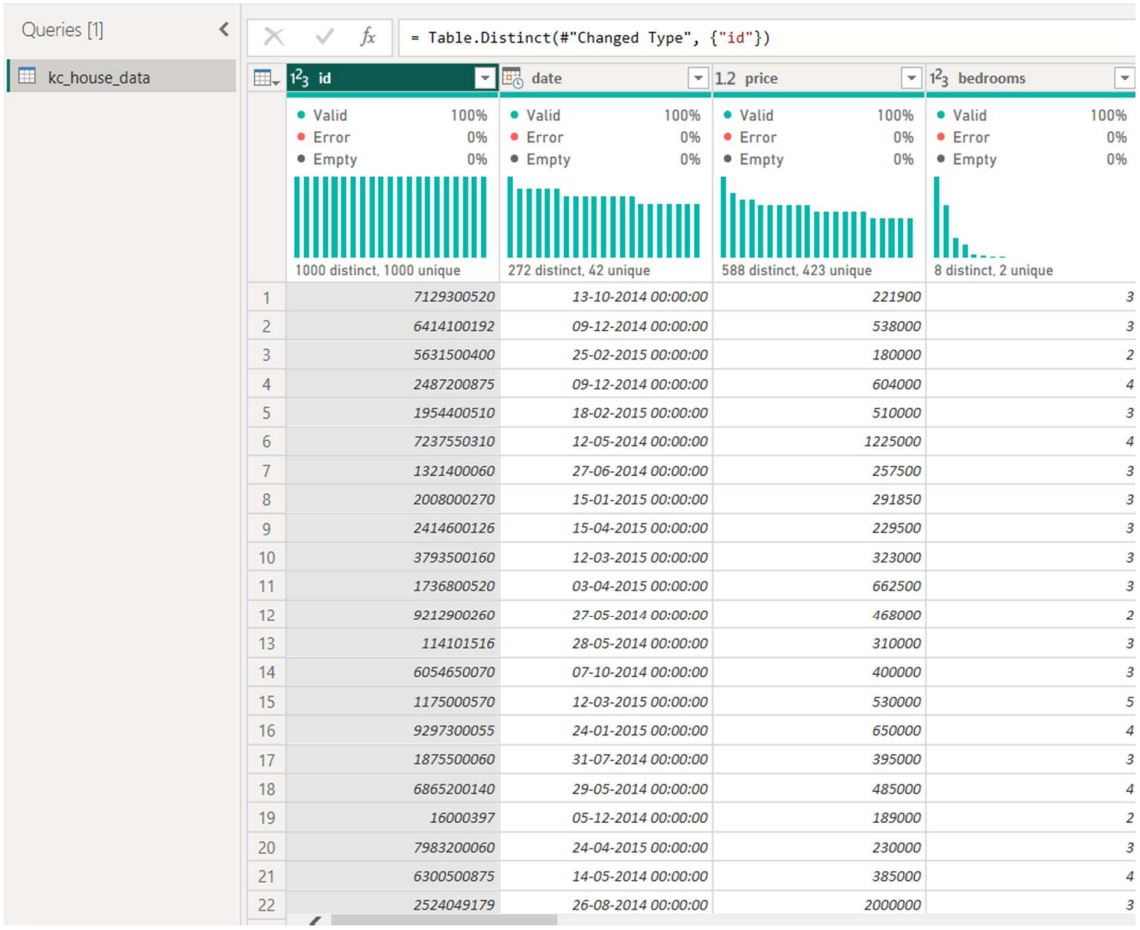


Figure 4: Checking column quality, missing values, and errors across the housing dataset in Power Query.

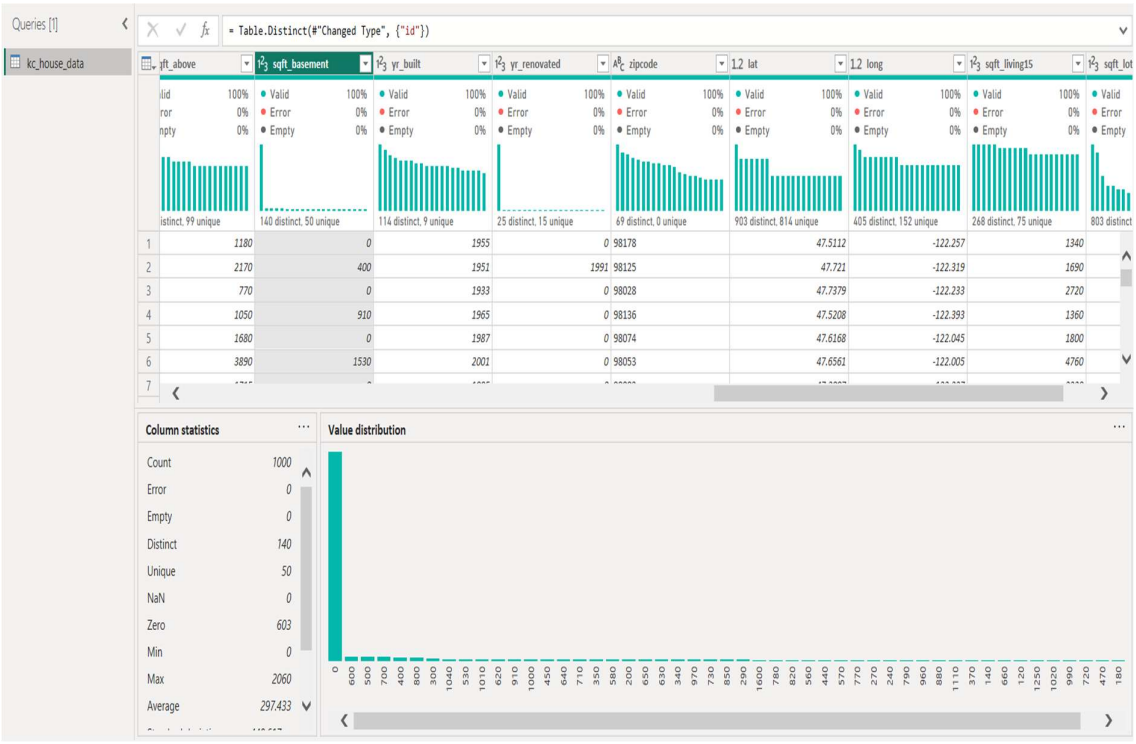
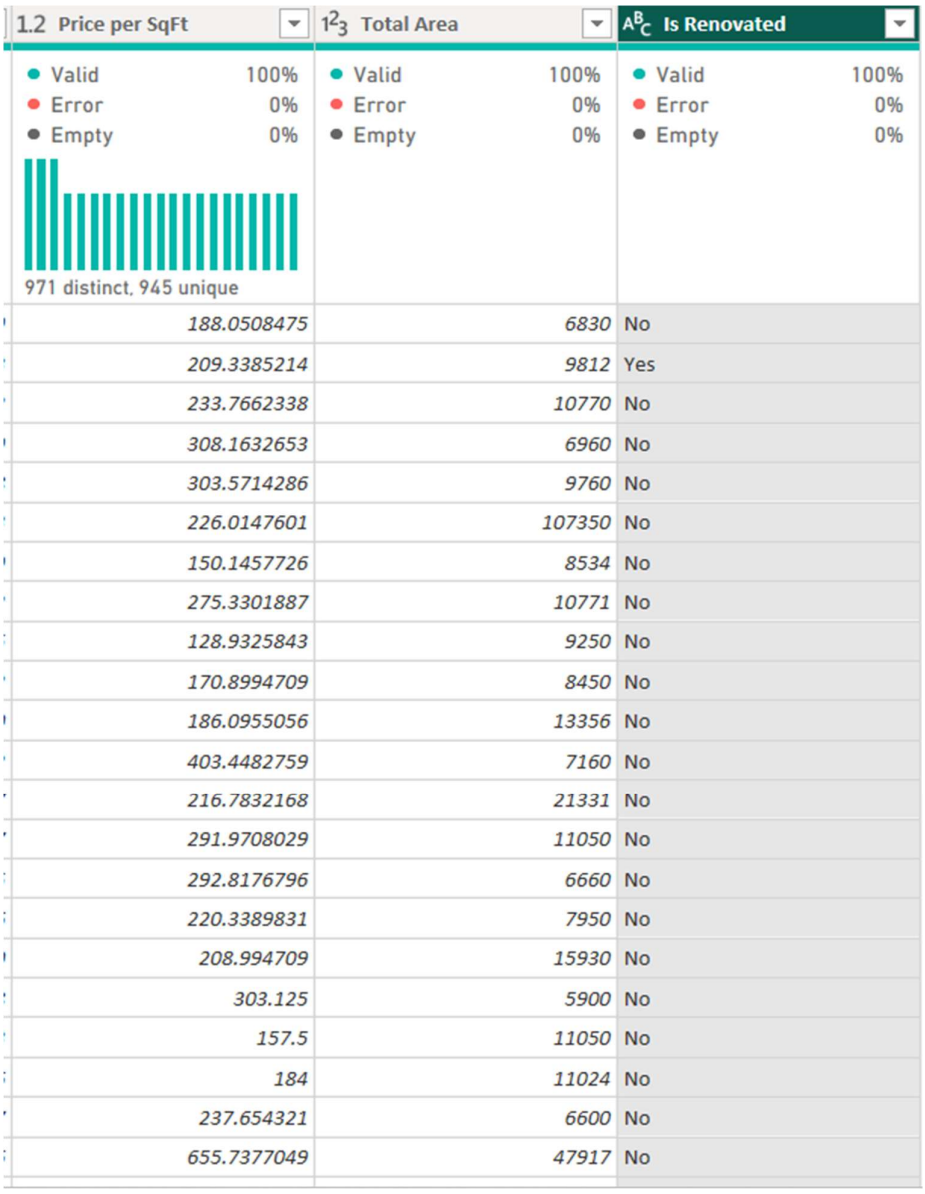


Figure 5: Extracting sale year and month and calculating property age at sale using Power Query custom columns.

Sale Year	Sale Month	Property Age at Sale
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
2014	10	59
2014	12	63
2015	2	82
2014	12	49
2015	2	28
2014	5	13
2014	6	19
2015	1	52
2015	4	55
2015	3	12
2015	4	50
2014	5	72
2014	5	87
2014	10	37
2015	3	115
2015	1	36
2014	7	20
2014	5	98
2014	12	93
2015	4	46
2014	5	67
2014	8	46

Figure 6: Creating Price per SqFt, Total Area, and Is Renovated columns in Power Query.



1.2 Price per SqFt	123 Total Area	ABC Is Renovated
Valid 100% Error 0% Empty 0% 971 distinct, 945 unique	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
188.0508475	6830	No
209.3385214	9812	Yes
233.7662338	10770	No
308.1632653	6960	No
303.5714286	9760	No
226.0147601	107350	No
150.1457726	8534	No
275.3301887	10771	No
128.9325843	9250	No
170.8994709	8450	No
186.0955056	13356	No
403.4482759	7160	No
216.7832168	21331	No
291.9708029	11050	No
292.8176796	6660	No
220.3389831	7950	No
208.994709	15930	No
303.125	5900	No
157.5	11050	No
184	11024	No
237.654321	6600	No
655.7377049	47917	No

Figure 7: Filtering invalid zero-bedroom records while preserving valid bathroom data in Power Query.

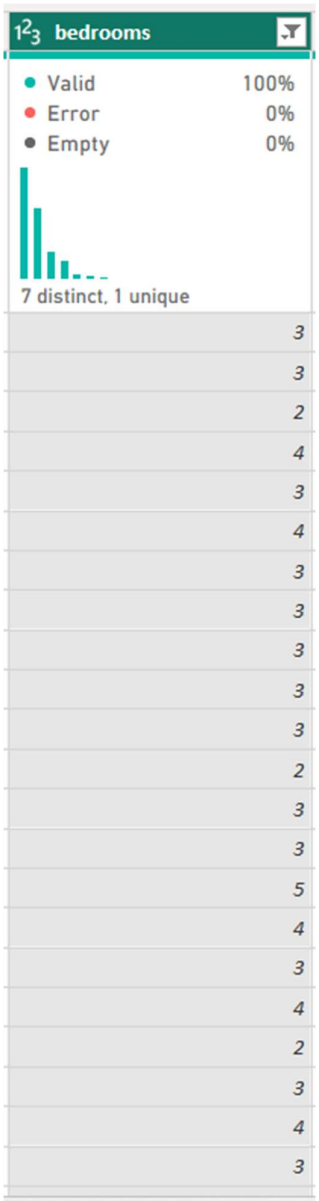
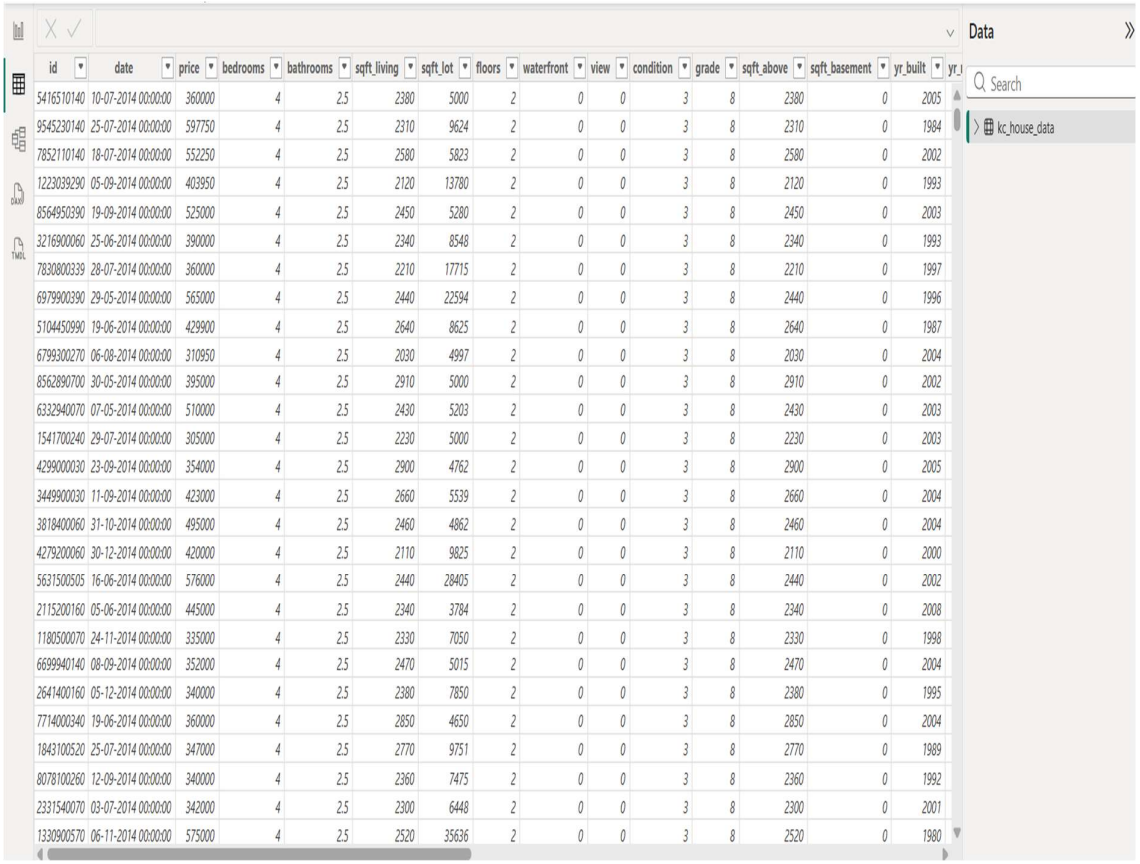


Figure 8: Categorizing properties by living area and sale price using Power Query conditional logic.

Property Size Category	Price Category
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
Medium	Low
Large	Medium
Small	Low
Medium	High
Medium	Medium
Very Large	Luxury
Medium	Low
Medium	Low
Medium	Low
Medium	Medium
Very Large	High
Medium	Medium
Medium	Medium
Medium	Medium
Medium	Medium
Large	High
Medium	Medium
Medium	Medium
Medium	Low
Medium	Low
Medium	Medium
Very Large	Luxury

Figure 9: Applying the cleaned and transformed housing dataset in Power BI for further analysis.



id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	grade	sqft_above	sqft_basement	yr_built	yr_
5416510140	10-07-2014 00:00:00	360000	4	2.5	2380	5000	2	0	0	3	8	2380	0	2005	
9545230140	25-07-2014 00:00:00	597750	4	2.5	2310	9624	2	0	0	3	8	2310	0	1984	
7852110140	18-07-2014 00:00:00	552250	4	2.5	2580	5823	2	0	0	3	8	2580	0	2002	
1223039290	05-09-2014 00:00:00	403950	4	2.5	2120	13780	2	0	0	3	8	2120	0	1993	
8564950390	19-09-2014 00:00:00	525000	4	2.5	2450	5280	2	0	0	3	8	2450	0	2003	
3216900060	25-06-2014 00:00:00	390000	4	2.5	2340	8548	2	0	0	3	8	2340	0	1993	
7830800339	28-07-2014 00:00:00	360000	4	2.5	2210	17715	2	0	0	3	8	2210	0	1997	
6979900390	29-05-2014 00:00:00	565000	4	2.5	2440	22594	2	0	0	3	8	2440	0	1996	
5104450990	19-06-2014 00:00:00	429900	4	2.5	2640	8625	2	0	0	3	8	2640	0	1987	
6799300270	06-08-2014 00:00:00	310950	4	2.5	2030	4997	2	0	0	3	8	2030	0	2004	
8562890700	30-05-2014 00:00:00	395000	4	2.5	2910	5000	2	0	0	3	8	2910	0	2002	
6332940070	07-05-2014 00:00:00	510000	4	2.5	2430	5203	2	0	0	3	8	2430	0	2003	
1541700240	29-07-2014 00:00:00	305000	4	2.5	2230	5000	2	0	0	3	8	2230	0	2003	
4299000030	23-09-2014 00:00:00	354000	4	2.5	2900	4762	2	0	0	3	8	2900	0	2005	
3449900030	11-09-2014 00:00:00	423000	4	2.5	2660	5539	2	0	0	3	8	2660	0	2004	
3818400060	31-10-2014 00:00:00	495000	4	2.5	2460	4862	2	0	0	3	8	2460	0	2004	
4279200060	30-12-2014 00:00:00	420000	4	2.5	2110	9825	2	0	0	3	8	2110	0	2000	
5631500505	16-06-2014 00:00:00	576000	4	2.5	2440	28405	2	0	0	3	8	2440	0	2002	
2115200160	05-06-2014 00:00:00	445000	4	2.5	2340	3784	2	0	0	3	8	2340	0	2008	
1180500070	24-11-2014 00:00:00	335000	4	2.5	2330	7050	2	0	0	3	8	2330	0	1998	
6699940140	08-09-2014 00:00:00	352000	4	2.5	2470	5015	2	0	0	3	8	2470	0	2004	
2641400160	05-12-2014 00:00:00	340000	4	2.5	2380	7850	2	0	0	3	8	2380	0	1995	
7714000340	19-06-2014 00:00:00	360000	4	2.5	2850	4650	2	0	0	3	8	2850	0	2004	
1843100520	25-07-2014 00:00:00	347000	4	2.5	2770	9751	2	0	0	3	8	2770	0	1989	
8078100260	12-09-2014 00:00:00	340000	4	2.5	2360	7475	2	0	0	3	8	2360	0	1992	
2331540070	03-07-2014 00:00:00	342000	4	2.5	2300	6448	2	0	0	3	8	2300	0	2001	
1330900570	06-11-2014 00:00:00	575000	4	2.5	2520	35636	2	0	0	3	8	2520	0	1980	

Figure 10: Creating DAX measures for average house price, living area, bedrooms, and bathrooms in Power BI.

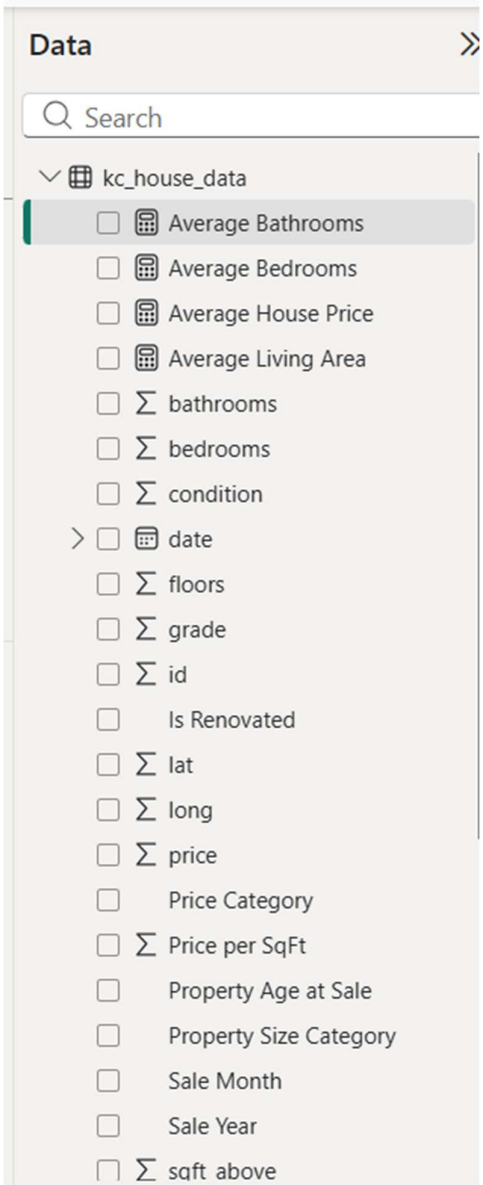


Figure 11: Creating total sales, median price, maximum price, and minimum price measures for housing analysis in Power BI.

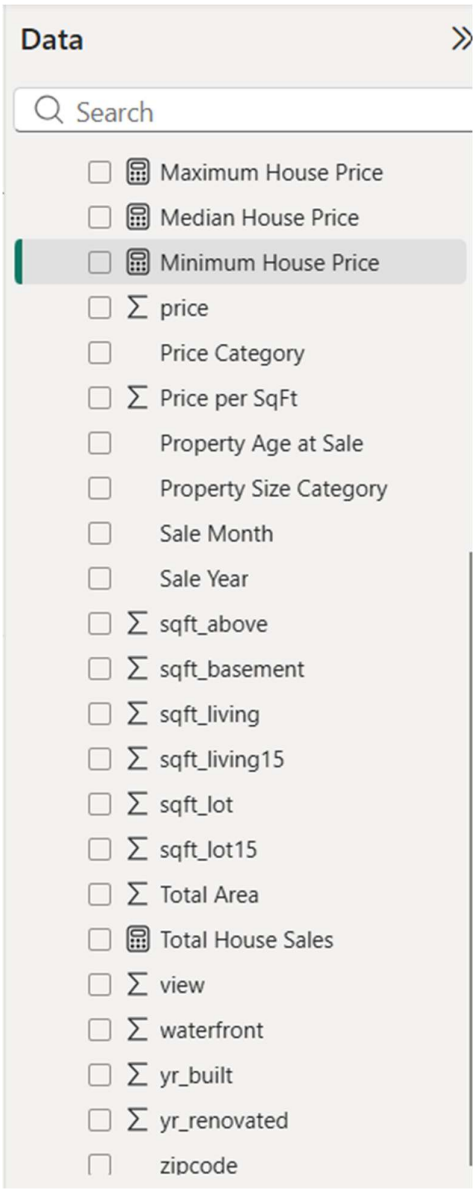


Figure 12: Creating additional DAX measures to analyze price per square foot, property age, and renovated property count.

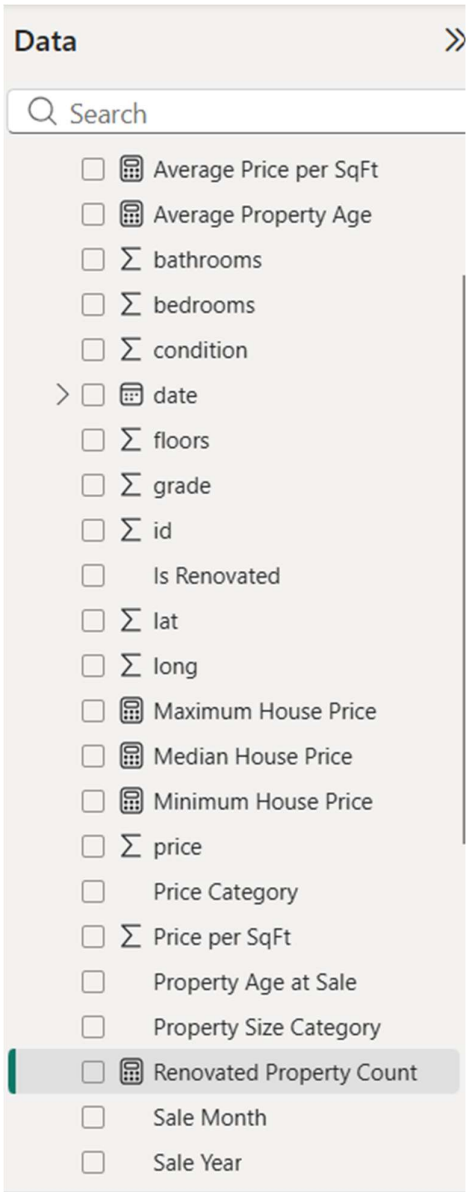


Figure 13: Validating housing analysis DAX measures using KPI card visuals in Power BI.

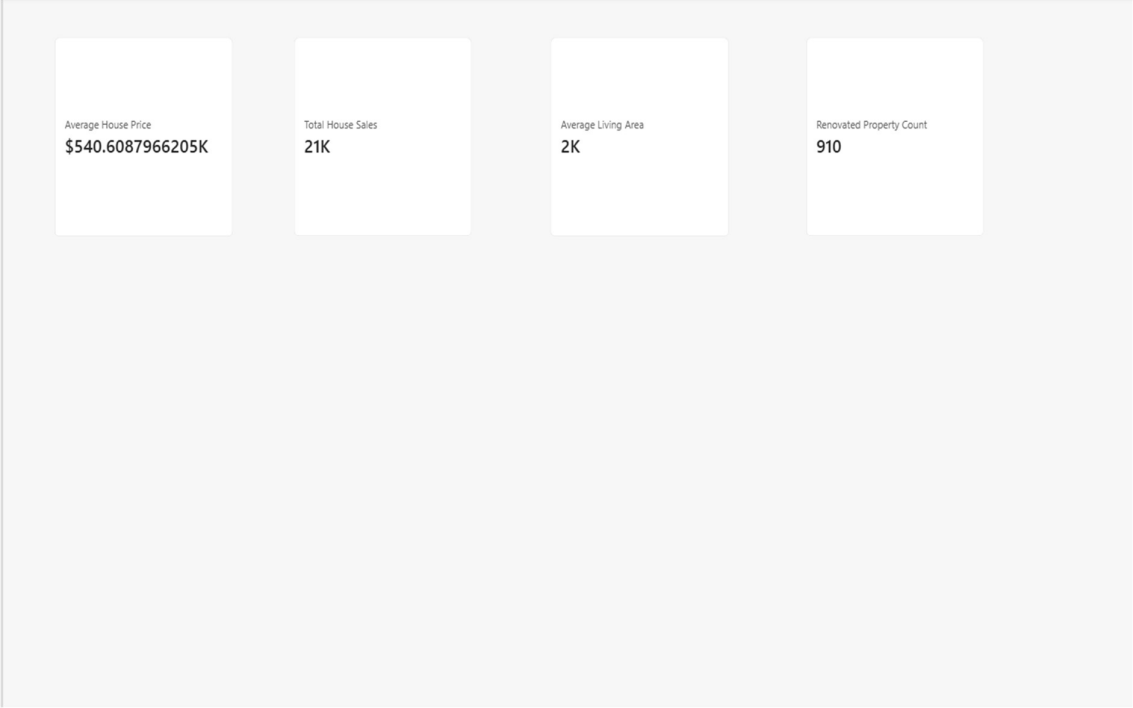
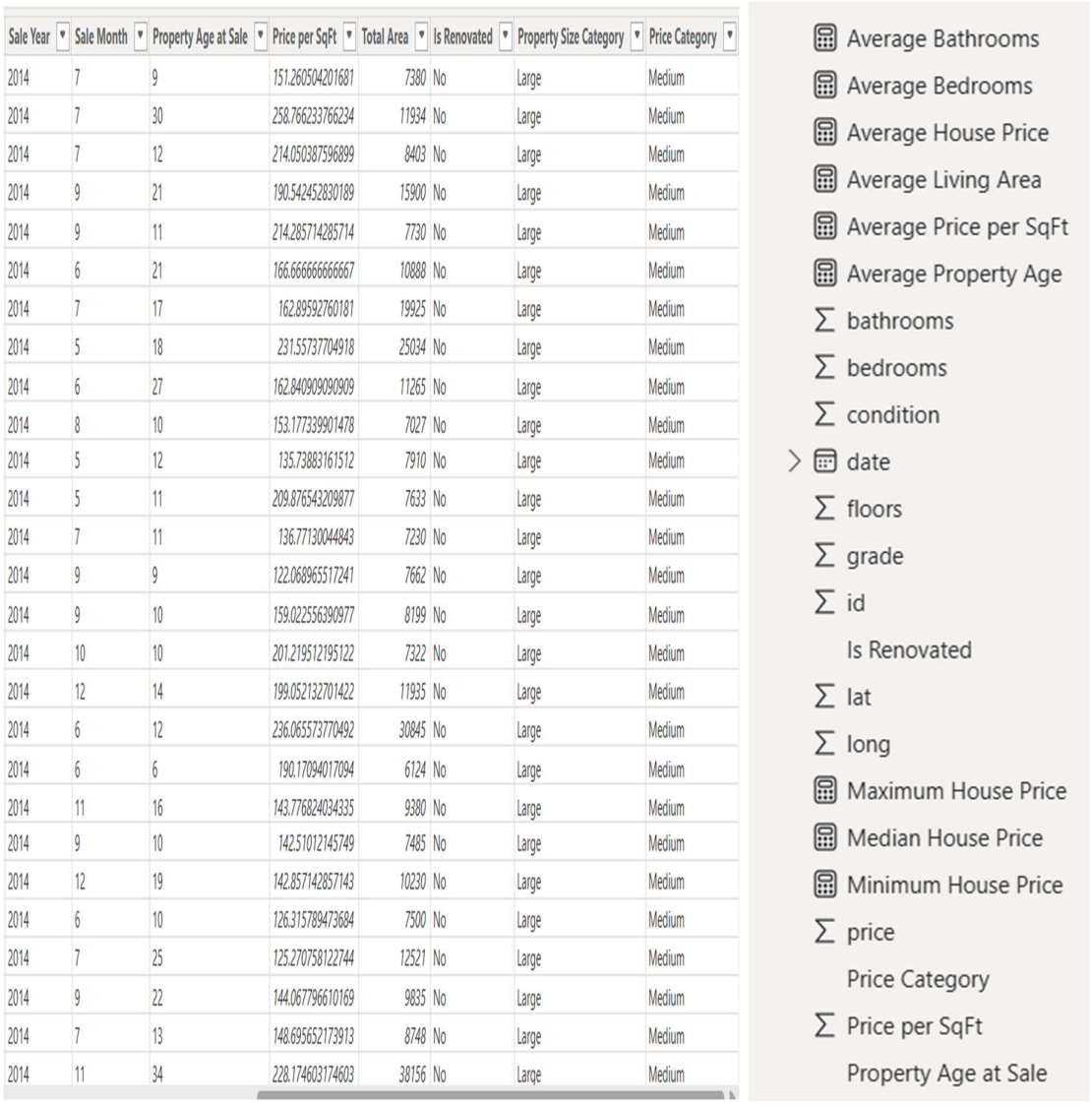


Figure 14: Validating the transformed housing dataset, calculated columns, and DAX measures in Power BI Table view.



Sale Year	Sale Month	Property Age at Sale	Price per SqFt	Total Area	Is Renovated	Property Size Category	Price Category
2014	7	9	151.260504201681	7380	No	Large	Medium
2014	7	30	258.766233766234	11934	No	Large	Medium
2014	7	12	214.050387596899	8403	No	Large	Medium
2014	9	21	190.542452830189	15900	No	Large	Medium
2014	9	11	214.285714285714	7730	No	Large	Medium
2014	6	21	166.666666666667	10888	No	Large	Medium
2014	7	17	162.89592760181	19925	No	Large	Medium
2014	5	18	231.55737704918	25034	No	Large	Medium
2014	6	27	162.840909090909	11265	No	Large	Medium
2014	8	10	153.177339901478	7027	No	Large	Medium
2014	5	12	135.73883161512	7910	No	Large	Medium
2014	5	11	209.876543209877	7633	No	Large	Medium
2014	7	11	136.77130044843	7230	No	Large	Medium
2014	9	9	122.068965517241	7662	No	Large	Medium
2014	9	10	159.022556390977	8199	No	Large	Medium
2014	10	10	201.219512195122	7322	No	Large	Medium
2014	12	14	199.052132701422	11935	No	Large	Medium
2014	6	12	236.065573770492	30845	No	Large	Medium
2014	6	6	190.17094017094	6124	No	Large	Medium
2014	11	16	143.776824034335	9380	No	Large	Medium
2014	9	10	142.51012145749	7485	No	Large	Medium
2014	12	19	142.857142857143	10230	No	Large	Medium
2014	6	10	126.315789473684	7500	No	Large	Medium
2014	7	25	125.270758122744	12521	No	Large	Medium
2014	9	22	144.067796610169	9835	No	Large	Medium
2014	7	13	148.695652173913	8748	No	Large	Medium
2014	11	34	228.174603174603	38156	No	Large	Medium

Conclusion

The Week 9 activity successfully completed the cleaning and transformation of the King County House Sales dataset using Power BI. The dataset was inspected for data quality, and relevant data types were verified. New calculated columns were created, including Sale Year, Sale Month, Property Age at Sale, Price per SqFt, Total Area, Is Renovated, Property Size Category, and Price Category.

DAX measures were also created to calculate average house price, total house sales, median house price, maximum and minimum house prices, average living area, average bedrooms, average bathrooms, average price per square foot, average property age, and renovated property count.

Finally, the transformed dataset and measures were validated to prepare the data for further analysis.

Overall, this activity provided practical experience in data cleaning, transformation, calculated columns, DAX measures, and data validation, preparing the housing dataset for the development of an interactive Real Estate Analytics Dashboard in Week 10.